Standard Operating Procedure for Investigating and Resolving Hardware/Environmental Faults in Ericsson PE Node ELEC-PE-06 at Electronic City Hub

Node: ELEC-PE-06 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_2444

Description

The Electronic City Hub, a critical telecom node in the network infrastructure, hosts Ericsson's PE node ELEC-PE-06, which is currently experiencing a hardware/environmental fault. The fault, identified by alarm code ALARM_2444, is characterized by ASIC memory parity errors and line card crashes, leading to a redundancy loss in the power supply. This SOP aims to provide a step-by-step guide for investigating and resolving this critical failure domain, considering the specific hardware and protocol mechanics of the Ericsson PE node. The PE node's architecture is based on a distributed system, where multiple line cards are connected through a high-speed interconnect, and each line card is equipped with a redundant power supply. The memory parity errors and line card crashes can be attributed to a variety of factors, including overheating, power supply fluctuations, or software bugs. In this context, the investigation and resolution of this fault require a thorough understanding of the PE node's hardware and software components, as well as the network protocols that govern data transmission between the line cards. This SOP will guide the technician through the diagnostic and resolution processes, ensuring that the fault is accurately identified and rectified in a timely and efficient manner.

Pre-requisite Checks

- Verify physical access to the Electronic City Hub and the Ericsson PE node ELEC-PE-06.

- Ensure that the network is in a stable state, and no other critical faults are present.

- Confirm that the redundant link status is operational, and the link is not experiencing any issues.

- Dispatch remote hands to the Electronic City Hub, if necessary, to assist with physical checks or equipment replacement.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	High
Power supply redundancy loss	Medium
Overheating	Low
Power supply fluctuations	Low
Software bugs	Low

Diagnostic Commands & Expected Outputs

```
Command: show controllers linecard, Expected Output: Line card status: OK Redundant link status: OK, Description: This command is used to verify the status of the line card and the redundant link. The expected output indicates that the line card is operational, and the redundant link is also functioning correctly.

Command: show memory parity errors, Expected Output: Memory parity errors: 10 Error rate: 0.01%, Description: This command is used to check for any memory parity errors on the PE node. The expected output indicates that there are 10 memory parity errors, with an error rate of 0.01%.

Command: show power supply status, Expected Output: Power supply status: OK Redundant power supply status: OK, Description: This command is used to verify the status of the power supply and the redundant power supply. The expected output indicates that both power supplies are operational.
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Resolution Steps

Step 1: Step: Replace the faulty line card, Command: replace linecard 1/1, Description: This step involves replacing the faulty line card with a new one. The command 'replace linecard 1/1' is used to initiate the replacement process.

Step 2: Step: Verify the redundant link status, Command: show controllers linecard, Description: After replacing the faulty line card, this step involves verifying the status of the redundant link. The command 'show controllers linecard' is used to check the status of the line card and the redundant link.

Step 3: Step: Reset the PE node, Command: reset node, Description: If the fault is still present after replacing the line card, this step involves resetting the PE node. The command 'reset node' is used to reset the node to its default configuration.

Post-Check Verification

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Command: show controllers linecard, Expected Output: Line card status: OK Redundant link status: OK, Description: This command is used to verify that the fault has been resolved. The expected output indicates that the line card is operational, and the redundant link is also functioning correctly.

Command: show memory parity errors, Expected Output: Memory parity errors: 0 Error rate: 0%, Description: This command is used to verify that the memory parity errors have been resolved. The expected output indicates that there are no memory parity errors.

Command: show power supply status, Expected Output: Power supply status: OK Redundant power supply status: OK, Description: This command is used to verify that the power supply and the redundant power supply are operational.
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Roll-back Procedures

- Step: Restore the original configuration, Command: restore config, Description: If the fix fails, this step involves restoring the original configuration of the PE node. The command 'restore config' is used to restore the original configuration.

- Step: Revert the changes, Command: revert changes, Description: This step involves reverting the changes made during the resolution process. The command 'revert changes' is used to revert the changes.